Course 6031C · Semester 6 · Program Elective · Revision 2021

Microcontroller & PLC

A professional engineering handbook for 8051 microcontroller architecture, assembly programming, peripheral interfacing, PLC hardware, ladder logic programming, PLC scan cycle and industrial control applications.

What this subject is

This course joins embedded control and industrial automation. The first half explains the 8051 microcontroller - CPU, memory, ports, registers, interrupts, serial communication, instruction set, assembly programming and 8255 PPI. The second half explains PLC - hardware blocks, scan cycle, inputs and outputs, ladder logic, timers, counters and industrial control programs.

Malayalam: 8051 microcontroller embedded control-□□ □□□□□□□□□□□□□□□□. PLC industrial control panel-□ relay logic □□□□□□□ programmable control □□□□□□□□□.

Official syllabus information

Item	Value
Program	Diploma in Electrical and Electronics Engineering
Course code	6031C
Course title	Microcontroller & PLC
Semester	6
Credits	4
Category	Program Elective
Periods	4 per week (L:3 T:1 P:0); 60 per semester

Course objectives

- Understand operation, construction and applications of microcontroller.
- Acquire basic knowledge in application and programming of 8051.
- Understand operation, construction and applications of PLC.
- Provide basic knowledge in application and programming of PLC.

Prerequisites

- Digital Electronics from Analog & Digital Circuits - Semester 3.
- Basics of programming from Introduction of IT Systems - Semester 1.

Course outcomes

- CO1: Summarize architecture of 8051 microcontroller.
- CO2: Develop programs to interface 8051 with peripherals.
- CO3: Illustrate PLC features, block diagram and applications.
- CO4: Build PLC programs for internal and external peripherals.

How to study

First master the difference between microprocessor, microcontroller and PLC. Then learn the 8051 internal blocks and memory map. After that write small assembly programs and understand interfacing diagrams. For PLC, learn the scan cycle, ladder symbols and practical circuits such as staircase lighting, go-down lighting, traffic light and DOL/star-delta starter.

Redesigned syllabus roadmap

Official module outcomes converted into a practical learning and exam route.

Module 1 · CO1 · 13 Hours

Architecture of 8051 Microcontroller

Summarize the features, block diagram, register structure, memory organization, pin diagram, interrupts, serial communication and applications of 8051.

Microcontroller vs microprocessor

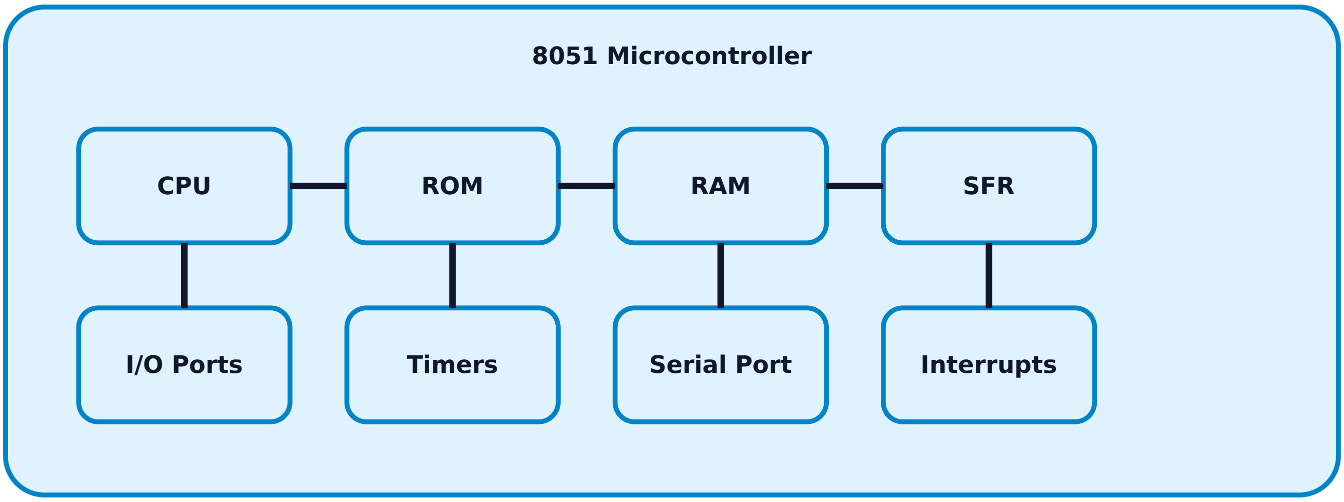
A microprocessor mainly contains CPU and needs external memory and I/O chips. A microcontroller integrates CPU, ROM, RAM, I/O ports, timers, serial port and interrupt controller on one chip. This makes it suitable for embedded control such as appliance controllers, sensor interfaces and simple machines.

Malayalam: Microprocessor-□□ external memory/I-O □□□□. Microcontroller-□ CPU, memory, ports, timer □□□□□□ □□□ chip-□ □□□□□□□□□ control applications-□□ □□□□□□□□.

Point	Microprocessor	Microcontroller
Integration	CPU centered	CPU + memory + I/O

Use	General computing	Dedicated control
Cost/size	More external chips	Compact system
Example	8085/8086 system	8051 based controller

8051 block diagram



Register structure

8051 has general purpose registers organized into register banks and special function registers (SFRs). Important SFRs include Accumulator A, B register, PSW, SP, DPTR, P0-P3, TMOD, TCON, IE, IP and SCON. PSW stores flag bits such as carry, auxiliary carry, overflow, parity and register bank selection bits.

> **Exam tip:** For PSW/TMOD/TCON/IE/IP questions, draw bit format first and then write the function of each bit in one line.

Memory organisation and pins

8051 has internal ROM and RAM and can access external program and data memory. Harvard style separation of program and data space is important. Ports P0 to P3 act as I/O; some pins have alternate functions such as serial communication, interrupts, timer inputs and external memory signals.

- P0: multiplexed address/data during external memory access.
- P1: general purpose I/O.
- P2: higher address bus during external memory access.
- P3: I/O plus RXD, TXD, INT0, INT1, T0, T1, WR and RD.

Interrupts and serial communication

Interrupts allow CPU to respond to urgent events without continuous polling. 8051 has external interrupts, timer interrupts and serial interrupt. IE register enables interrupts; IP controls priority. Serial communication uses TXD and RXD pins and SCON register for elementary serial control.

Applications

- Motor and relay control
- LED display and keypad systems
- Small appliance control
- Embedded measurement instruments
- Stepper motor control
- Industrial timers and counters

Module 2 · CO2 · 17 Hours

8051 Programming and Interfacing

Develop simple assembly programs and interface 8051 with internal and external peripherals such as LED, motors and 8255 PPI.

Instruction format and addressing modes

An 8051 instruction has operation code and operand fields. Addressing mode decides where the operand is located. Important modes are immediate, register, direct, indirect, indexed and bit addressing.

Mode	Example	Meaning
Immediate	MOV A,#25H	Data is inside instruction.
Register	MOV A,R0	Operand in register.
Direct	MOV A,30H	Internal RAM/SFR address.
Indirect	MOV A,@R0	Register holds address.
Bit	SETB P1.0	Single bit operation.

Instruction set groups

- **Data transfer:** MOV, MOVC, MOVX, PUSH, POP.
- **Arithmetic:** ADD, ADDC, SUBB, INC, DEC, MUL, DIV.
- **Logical:** ANL, ORL, XRL, CPL, CLR, RL, RR.
- **Branch/control:** SJMP, LJMP, ACALL, RET, JZ, JNZ, CJNE, DJNZ.

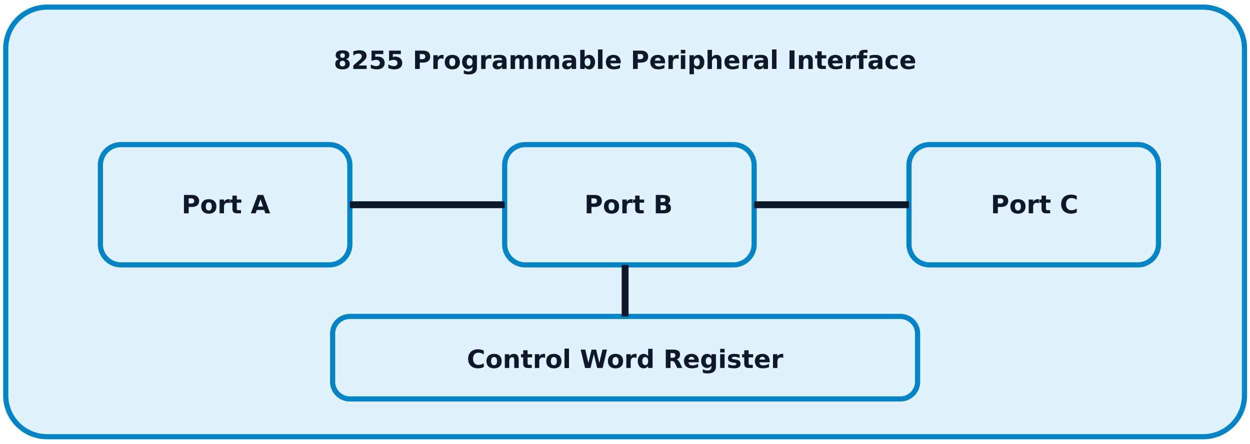
Malayalam: ALP □□□□□□□□□□ data □□□□□ □□□□□□□□ □□□□□□□□, operation □□□□□□□, result □□□□□ □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□□□.

Simple ALP pattern

For exam programs, write initialization, main operation, result storage and stop/end loop. Always mention port direction assumption and comments.

```
; Add two numbers in internal RAM
MOV A,30H      ; first number
ADD A,31H      ; add second number
MOV 32H,A      ; store result
HERE: SJMP HERE
```

8255 PPI block diagram



Interfacing LED and motors

LED interfacing needs current limiting resistor and correct port logic. DC motor interfacing requires driver transistor, relay or H-bridge because port pins cannot directly supply motor current. Stepper motor interfacing energizes coils in a proper sequence through driver ICs.

> **Safety/practical note:** Never connect motor directly to 8051 port pins. Use a driver stage and flyback protection for inductive loads.

Expected questions

1. Explain addressing modes of 8051 with examples.
2. Write an ALP to add two numbers.
3. Draw 8255 PPI block diagram and control word register.
4. Draw connection diagram for LED, DC motor and stepper motor interfacing.

Module 3 · CO3 · 13 Hours

Programmable Logic Controller Architecture

Illustrate PLC features, block diagram, scan cycle, applications and selection criteria.

Importance of PLC

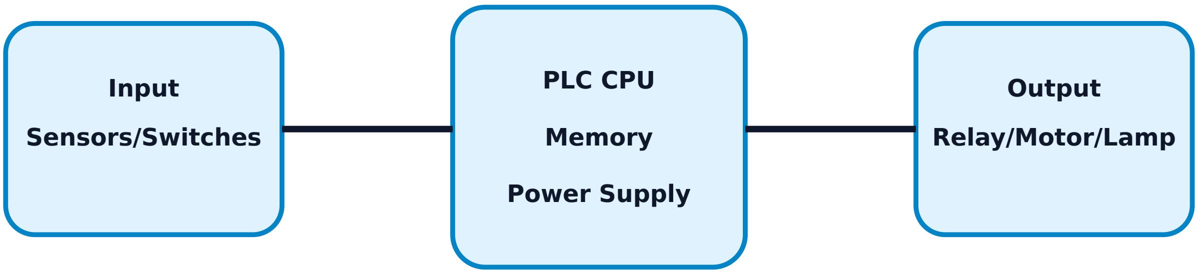
A PLC is an industrial computer designed for rugged real-time control. It reads input devices, executes stored logic and drives output devices. PLCs replaced many relay-based control panels because they are easier to modify, diagnose and expand.

Malayalam: Relay panel-□ wiring □□□□□□ logic □□□□□□□. PLC-□ program change □□□□□□□ control logic □□□□□□□ □□□□□□.

Relay panel vs PLC panel

Point	Conventional relay panel	PLC based panel
Logic change	Requires rewiring	Program modification
Fault finding	Time consuming	Diagnostics and online status
Size	Large for complex logic	Compact
Expansion	Difficult	I/O modules can be added
Reliability	Mechanical wear	High, fewer moving parts

PLC based system



PLC operation and scan cycle

The PLC scan cycle has four main stages: read inputs, execute program, update outputs and perform communication/diagnostics. Fast and repeated scanning makes PLC suitable for industrial automation.

1. Input scan - read sensors and switches.
2. Program scan - execute ladder logic.
3. Output scan - energize output modules.
4. Housekeeping - diagnostics and communication.

Applications and selection

PLC is used in conveyor control, pump control, escalator/elevator auxiliary logic, machine sequence control, packaging machines, traffic lights and motor starters. Selection depends on number of I/O, voltage level, memory, scan time, communication, environment, expansion and cost.

Common mistakes

- Ignoring input/output voltage rating.
- Not providing interlocking for motors.
- Using PLC output directly for high current loads without contactor/relay.
- Not considering future spare I/O.

Module 4 · CO4 · 15 Hours

PLC Programming and Ladder Logic

Build ladder programs for logic gates, lighting circuits, traffic light, DOL starter and star-delta starter.

Basics of PLC programming

PLC programs may be written using ladder diagram, function block diagram, instruction list, structured text and sequential function chart. The syllabus focuses on ladder logic and standard PLC instruction types.

Ladder logic rules

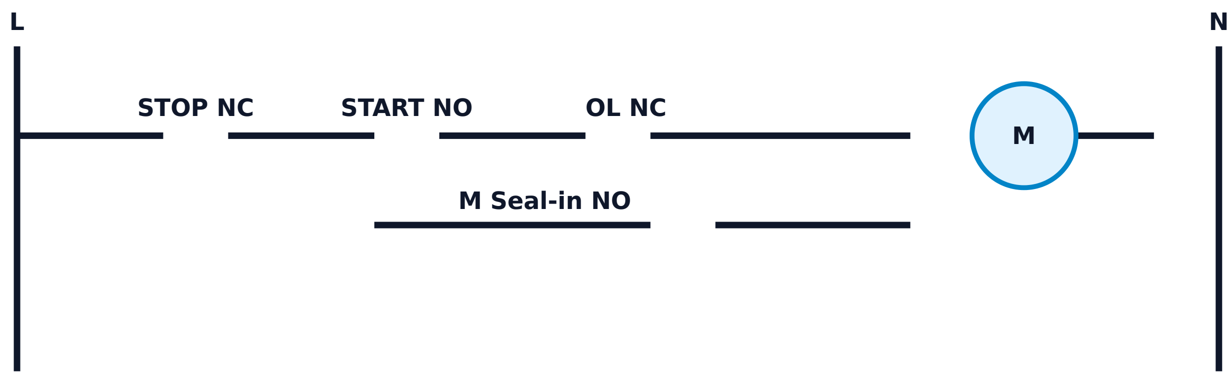
- Power rails are drawn on left and right.
- Each rung represents one control logic path.
- Inputs/contacts are placed before outputs/coils.
- Output coil should normally appear at the right end of the rung.
- Seal-in contact is used for latching starter circuits.
- Interlocks prevent unsafe simultaneous operation.

Malayalam: Ladder logic-□ rung □□□ control condition □□□. Input contact true □□□□□□□□ output coil energize □□□□□□□.

Instruction sets

- Basic I/O instructions: NO contact, NC contact and coil.
- Timer/counter instructions: TON, TOF, CTU, CTD.
- Math instructions: add, subtract, multiply, divide.
- Logical instructions: AND, OR, NOT.
- Data handling: move, compare and register operations.

DOL starter ladder concept



Programs required by syllabus

- Logic gates using ladder contacts.
- Staircase lighting circuit.
- Go-down lighting circuit.
- Traffic light sequence.
- DOL motor starter.

- Star-delta motor starter.

Exam answer-writing format

For ladder program questions, write I/O address table first, draw ladder diagram, explain rung-by-rung operation, then mention safety interlock/timer requirement.

Formula / Rule / Instruction Bank

Programming subjects need rules, instruction meanings, timing ideas and diagram checklists more than formulas.

8051 instruction format

Instruction = Opcode + Operand(s). Opcode says operation; operand says data/source/destination.

Addressing rule

Identify where the data is: immediate data, register, direct address, indirect address, indexed table or bit.

ALP rule

Initialize -> perform operation -> store result -> provide stop/end loop.

PLC scan rule

Read inputs -> execute program -> update outputs -> diagnostics/communication.

Ladder rule

Inputs/conditions on left, output coil on right, one main output per rung where possible.

Timer rule

Use ON delay for delayed starting; use OFF delay for delayed stopping.

Motor control rule

Always include stop, overload and interlock logic in motor starter ladders.

Interface rule

Do not connect high current load directly to microcontroller or PLC transistor output; use driver/relay/contactor.

Solved Examples / Problem Lab

Step-by-step solutions and program patterns.

Example 1 - Add two 8-bit numbers in 8051

Given: first number in 30H and second number in 31H.

Logic: move first number to accumulator, add second number, store result.

```
MOV A,30H
ADD A,31H
MOV 32H,A
HERE: SJMP HERE
```

Result stored in 32H.

Example 2 - LED ON at P1.0

Use bit operation on Port 1 bit 0. Assume active high LED through resistor.

```
SETB P1.0
HERE: SJMP HERE
```

P1.0 becomes logic 1 and LED turns ON.

Example 3 - PLC AND gate

Inputs: X0 and X1. Output: Y0.

Ladder: X0 NO contact in series with X1 NO contact drives Y0 coil.

Y0 ON only when X0=1 and X1=1.

Example 4 - DOL starter logic

Inputs: STOP NC, START NO, OL NC. Output: motor contactor M.

Use M auxiliary NO contact parallel to START for seal-in/latching.

Press START -> M ON and latched. Press STOP or overload -> M OFF.

Example 5 - Traffic light sequence

Use timers T1, T2 and T3 for green, yellow and red intervals.

Logic cycles outputs: Green -> Yellow -> Red -> repeat.

Timer-based sequencing is expected, not random switching.

Practice Section and Expected Questions

Module-wise exam preparation.

One-mark questions

1. What is SFR?
2. Name any two 8051 ports.
3. What is PLC scan cycle?
4. What is ladder rung?
5. What is 8255 PPI?

Short-answer questions

1. Compare microprocessor and microcontroller.
2. List important SFRs of 8051.
3. Explain addressing modes with examples.
4. List advantages of PLC based panel.
5. Explain rules for writing ladder logic.

Descriptive questions

1. Draw and explain 8051 architecture.
2. Explain memory organization of 8051.
3. Draw 8255 PPI block diagram and explain control word register.
4. Draw and explain PLC block diagram and scan cycle.
5. Develop ladder program for DOL and star-delta starter.

MCQ practice

1. Which register contains 8051 flags? A) DPTR B) PSW C) TMOD D) SCON

2. PLC output is updated during which stage? A) Input scan B) Output scan C) Program edit D) Download

3. In ladder logic, contacts normally represent: A) Inputs/conditions B) Output coils C) Memory only D) Power supply only

Fresh Model Question Paper

Original question paper based on syllabus coverage; not copied from official question papers.

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Time: 3 Hours **Maximum Marks:** 100

Part A - Short Answer

1. Define microcontroller.
2. Name any two SFRs of 8051.
3. What is immediate addressing mode?
4. What is PLC scan cycle?
5. What is ladder logic?

Part B - Medium Answer

1. Compare microprocessor and microcontroller.
2. Explain the register structure of 8051.
3. Explain 8255 PPI block diagram.
4. Compare conventional relay panel and PLC based panel.
5. Explain rules for writing a ladder logic diagram.

Part C - Long Answer / Programs

1. Draw and explain the architecture of 8051 microcontroller.
2. Write 8051 ALP to add two numbers and store the result.
3. Explain PLC block diagram, I/O devices and scan cycle.
4. Develop ladder logic for DOL starter with START, STOP and overload protection.
5. Develop a ladder sequence for traffic light control using timers.

Complete Model Answers

Answer guide with keywords, diagrams to draw and common mistakes to avoid.

Microcontroller definition

A microcontroller is a single-chip control device containing CPU, memory, I/O ports, timers, serial port and interrupt logic for embedded applications.

8051 architecture answer points

Draw CPU, oscillator, ROM, RAM, SFR, I/O ports, timers/counters, serial port and interrupt control. Explain each block in 1-2 lines and mention embedded control applications.

ALP addition answer

```
MOV A,30H
ADD A,31H
MOV 32H,A
HERE: SJMP HERE
```

Common mistake: not storing the result or not showing the end loop.

PLC scan cycle answer

Read inputs, execute ladder program, update outputs and perform diagnostics/communication. The cycle repeats continuously.

DOL ladder answer

Input table: START NO, STOP NC, OL NC. Output: contactor M. Ladder has STOP and OL in series, START NO in series, M coil at right and M auxiliary NO seal-in contact parallel to START.

Relay vs PLC panel

Relay panels require wiring changes for logic changes, have more mechanical contacts and are harder to troubleshoot. PLC panels use programmable logic, diagnostics, less wiring, compact layout and easy modification.